Experiment with linear functions and relations using digital tools, making and testing conjectures and generalising emerging patterns

Learning intention

- experiment with linear functions and relations using digital tools, making and testing conjectures and generalising emerging patterns.

Teach and model

- using digital tools to investigate integer solutions to equations such as $2x+3y\;=\;48$.
- exploring how linear functions are used in linear regression models as a statistical technique in machine learning of artificial intelligence agents.
- using graphing software to systematically contrast the graphs of $y\;=\;2x$, $-y\;=\;2x$, $y\;=\;-2x$ and $-y\;=\;-2x$ with those of y 2x, $-y>2x$, $y>-2x$ and $-y>-2x$.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.